
What students need to learn

1.1 Healthy and Active Lifestyles

The development of active leisure and recreation

Requirements for participation

Students should understand the following concepts and be able to apply them to practical examples:

- fitness
- ability
- resources
- time

Concepts of recreation and active leisure

- Philosophy of mass participation, including its origins and the principles behind it
- Sport for all, including its history and the impact of related policies

Contemporary concerns

Students should understand the following contemporary concerns, including recognising their causes, related implications on health and how active and healthy lifestyles can reduce potential negative effects:

- obesity, coronary heart disease, diabetes, high blood pressure, high cholesterol, metabolic syndrome, sedentary lifestyles, activity patterns, ageing population, access, stress, negative lifestyle choices, including smoking, drinking and drugs.

Healthy lifestyle

Health, fitness and exercise

Students should understand the differences and links between health, fitness and exercise:

- physical (energy expenditure)
- psychological

Positive health benefits:

- reduction in body fat
- increased resting metabolic rate and/or increased proportion of muscle mass
- reduced rates of mortality, coronary heart disease (CHD), obesity, reduced risk of osteoporosis, and help type II diabetes management

Nutrition and weight management

Energy intake, food groups, hydration, balanced diet, food pyramid, cholesterol, dietary requirements for exercise

Current trends in health

Comparative cultures, eg USA, Finland, Japan

Balanced lifestyle

- Energy balance (energy intake versus energy expenditure)
- Work-life balance, including dealing with stress

Effects of ageing

Students should understand the point of physical maturation, how and why this differs between sports, and the effects of ageing there after, including:

- decline in physical capability (cardiovascular fitness, muscular strength, lung function, resting metabolic rate, osteoporosis, flexibility)

Effects of exercise

Responses and adaptations of the body systems

Students should understand why body systems respond and adapt as they do, and the differences between structural and functional responses and adaptations:

Muscular-skeletal system

- Responses (fibre recruitment, force production, metabolism)
- Adaptations (increase fibre size, increase force production, increase metabolic energy supply)

Cardiovascular system

- Responses (increase stroke volume, heart rate, cardiac output, redistribution of blood flow to working muscle)
- Adaptations (increase in size of heart, stroke volume, cardiac output and number of capillaries)

Respiratory system

- Responses (increased rate and depth of breathing, increased gas exchange)
- Adaptations (improvements in respiratory muscle performance)

Neuro-muscular system

- Responses (increased number of muscle fibres recruited, increase in rate of fibre recruitment)
- Adaptations (improved co-ordination, increases in force production and rate of force production, increase in speed)

Fitness and training

Components of physical fitness

Speed, strength, endurance (aerobic, muscular), flexibility, body composition

Components of skill fitness

Co-ordination, balance, agility, reaction time, power, strength, flexibility, endurance (local muscle, cardiovascular, strength)

Methods of fitness training

Aerobic/anaerobic training, continuous, interval, power, circuit/weight/resistance training, speed, fartlek, cross training, core stability, plyometrics, SAQ (speed, agility, quickness), stretching (static, ballistic, dynamic, proprioceptive neuromuscular facilitation — PNF)

Principles of training

- FITT (frequency, intensity, time, type)
- Application of fitness and training/overtraining
- SPORT (specificity, progression, overload, reversibility, time)

Appropriateness of programmes to differing clients

Young and old, trained and untrained, active and sedentary, healthy and unhealthy

Fitness assessment

Rationale of assessment

Maximal v sub-maximal tests, reasons for testing/use of tests for assessment of fitness, limitations, validity and reliability, using technology to assess fitness

Protocols of recognised fitness tests

Application, understanding, Physical Activity Readiness Questionnaire (PARQ), tests for different components of fitness, eg strength, power, speed, flexibility, endurance, body composition